

Lesson 15: Power and sample size calculations for means

TB sections 5.4

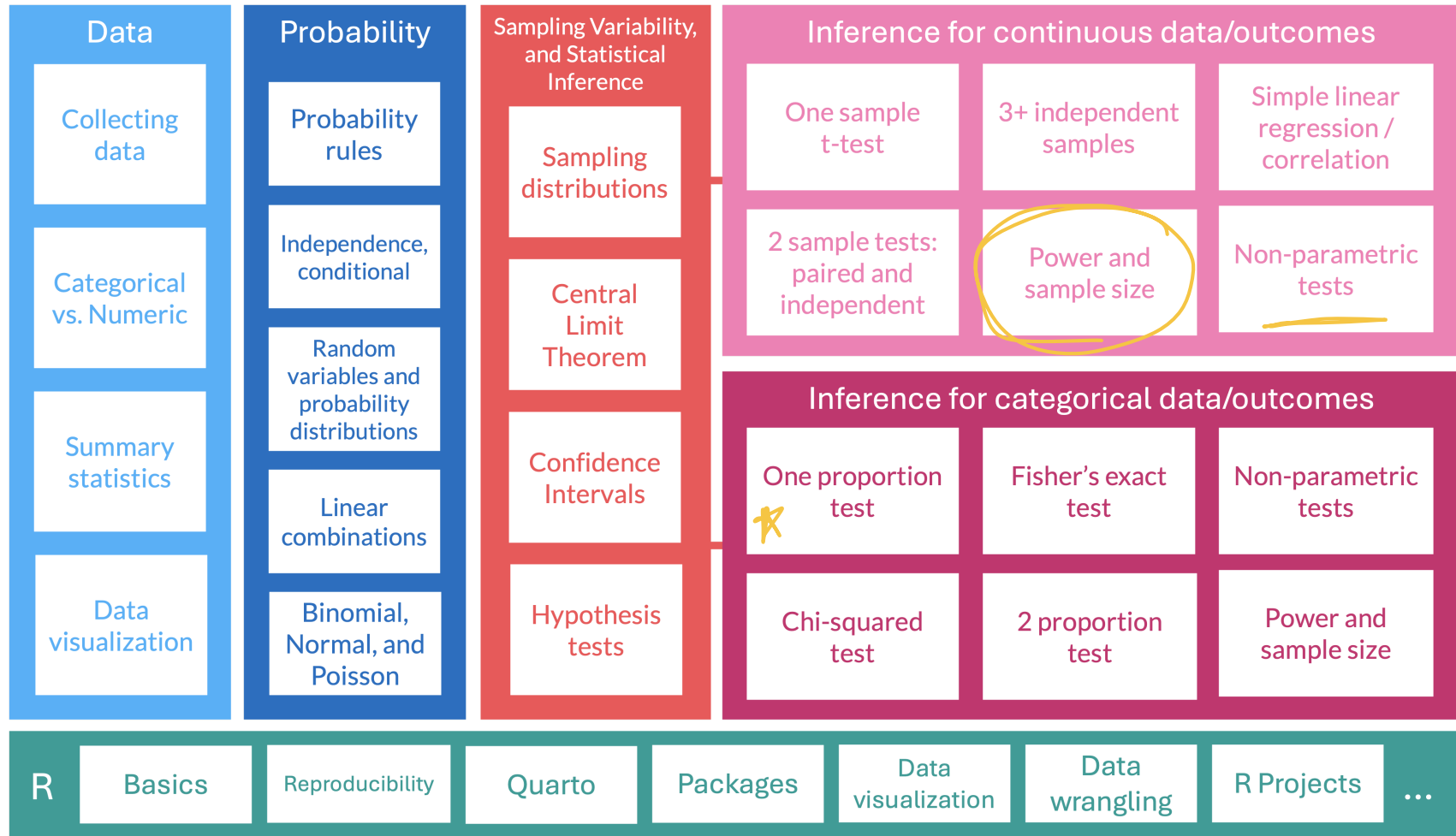
Meike Niederhausen and Nicky Wakim

2025-11-17

Learning Objectives

1. Understand the four components in equilibrium in a hypothesis test.
2. Define the significance level, critical value, and rejection region.
3. Define power and understand its role in a hypothesis test.
4. Understand how to calculate power for two independent samples.
5. Using R, calculate power and sample size for a single mean t-test and two independent mean t-test.

Where are we?



Before we get into power and sample size

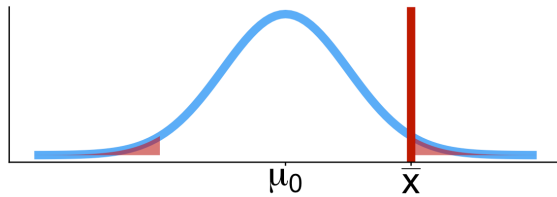
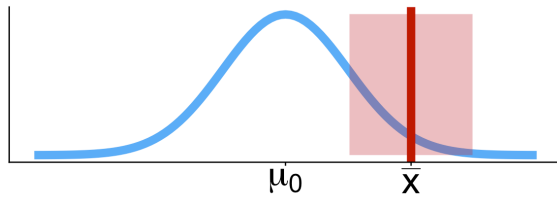
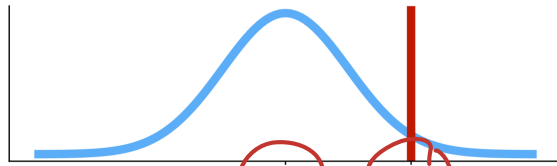
Let's watch this youtube video to remind us of what we learned (stop at 3:30):

One sample t-test vs Independent t-test vs Paired t-test

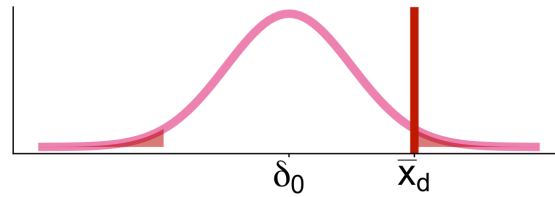
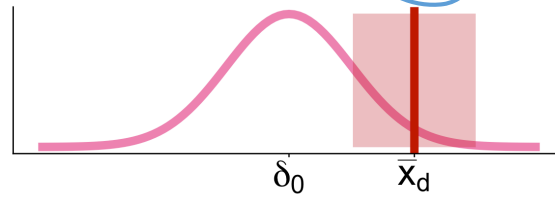
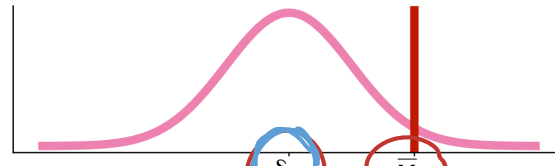


From Lesson 12-14: What do our hypothesis tests look like?

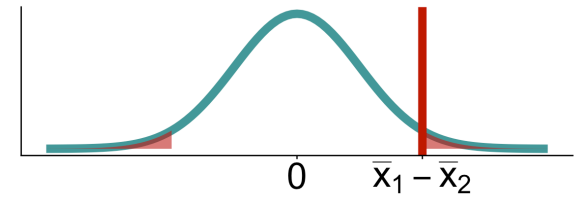
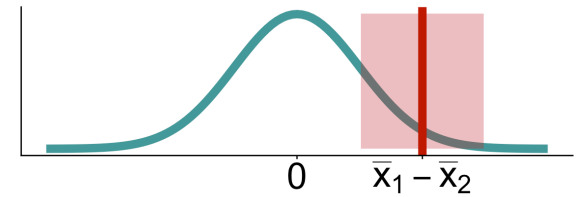
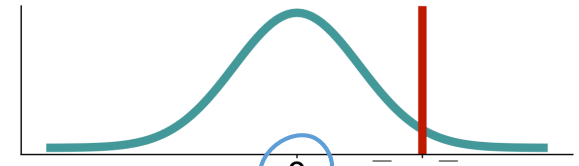
Single-sample mean:



Paired mean difference:



Diff in means of 2 ind samples:



From Lesson 12-14: What do our hypothesis tests look like?

Try filling out this table:

	One-sample	Independent two-sample	Paired sample
Example	Body temp: Population mean is 98.6°F, is sample different?	Caffeine: taps/min between caffeine and non-caffeine group	Vegetarian diet: cholesterol before and after
Sample statistic	$\bar{x}$	$\bar{x}_1 - \bar{x}_2$	$x_d = x_1 - x_2$ (for each person) & $\bar{x}_d$
Population parameter	μ	$\mu_1 - \mu_2$	δ
Possible hypothesis tests	$H_0: \mu = 98.6$ $H_A: \mu \neq 98.6$	$H_0: \mu_1 = \mu_2$ $H_A: \mu_1 > \mu_2$	$H_0: \delta = 0$ $H_A: \delta \neq 0$
Standard error if pop σ Not known	$s/\sqrt{n}$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$s_d/\sqrt{n}$ s_d
Test statistic			
Confidence intervals			

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From Lesson 14: Does caffeine increase finger taps/min (on average)?

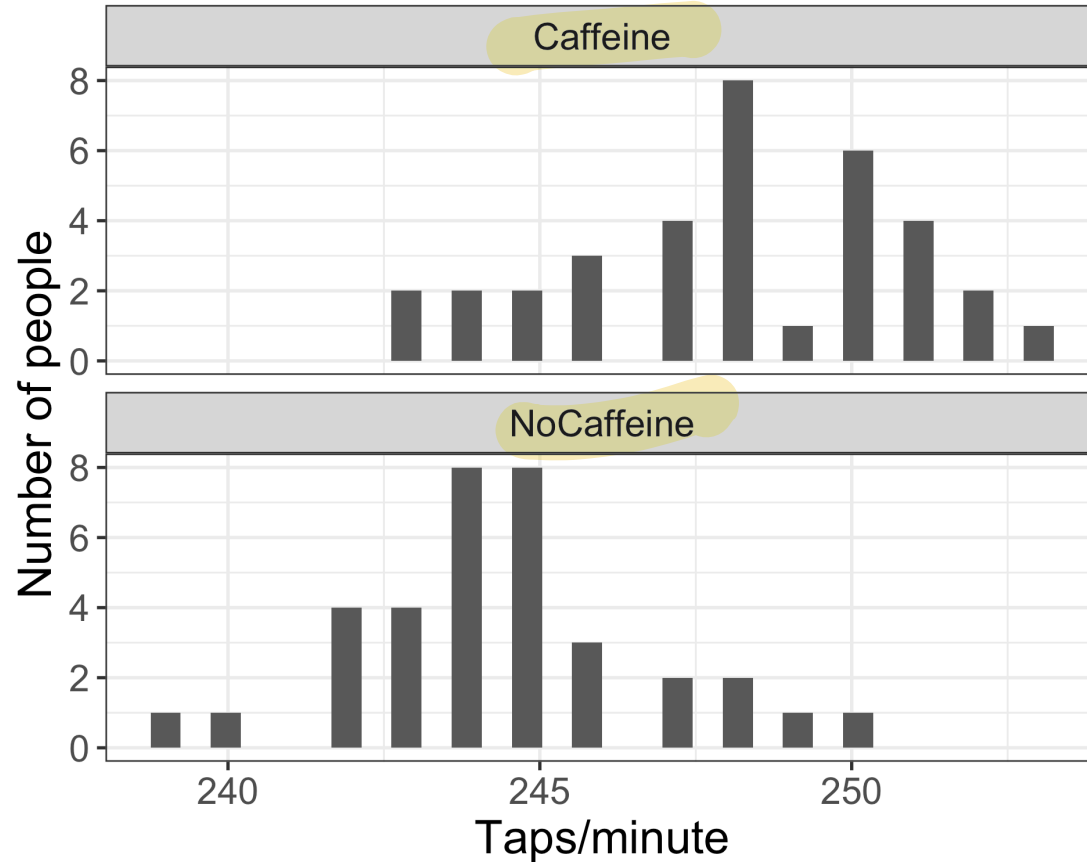
- Used this example to illustrate how to calculate a confidence interval and perform a hypothesis test for two independent samples

Study Design:¹

- 70 college students were trained to tap their fingers at a rapid rate
- Each then drank 2 cups of coffee (double-blind)
 - **Control** group: decaf
 - **Caffeine** group: ~ 200 mg caffeine
- After 2 hours, students were tested.
- **Taps/minute** recorded

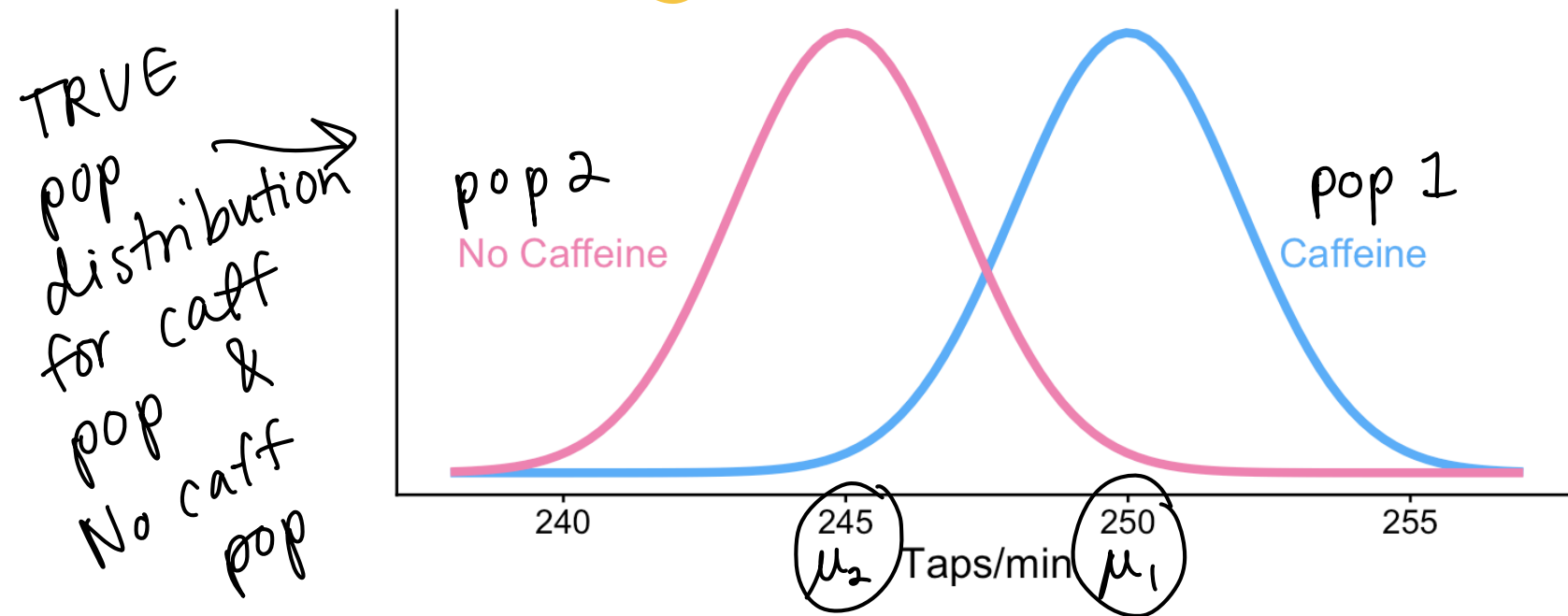
We started looking at the taps/min for each group

- Code to make these histograms



What if the following were the true population distributions? Case 1

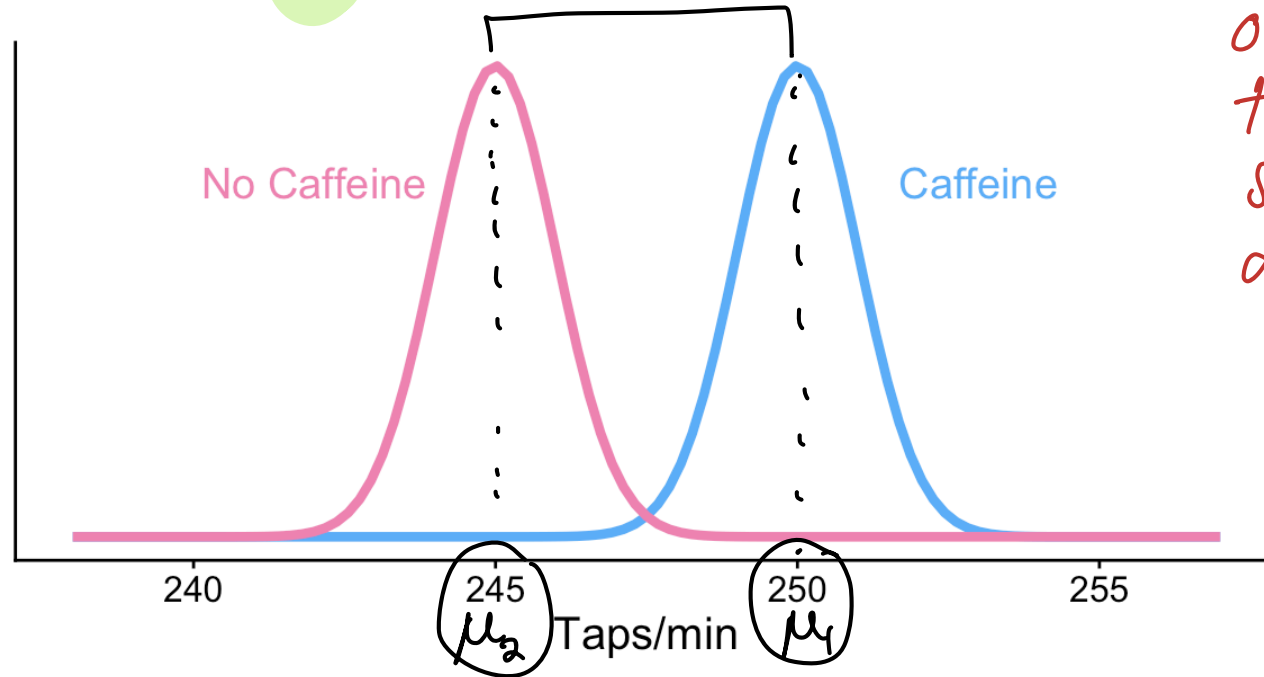
- Difference in population means is 5: caffeine has 5 taps/min more on average
- Both have a standard deviation of 2



- When we take two samples from these groups, do you think it would be easy to distinguish between the mean taps/min?
 - Depends on the number of samples we get: we might need a lot

What if the following were the true population distributions? Case 2

- Difference in population means is 5
- Both have a standard deviation of 1



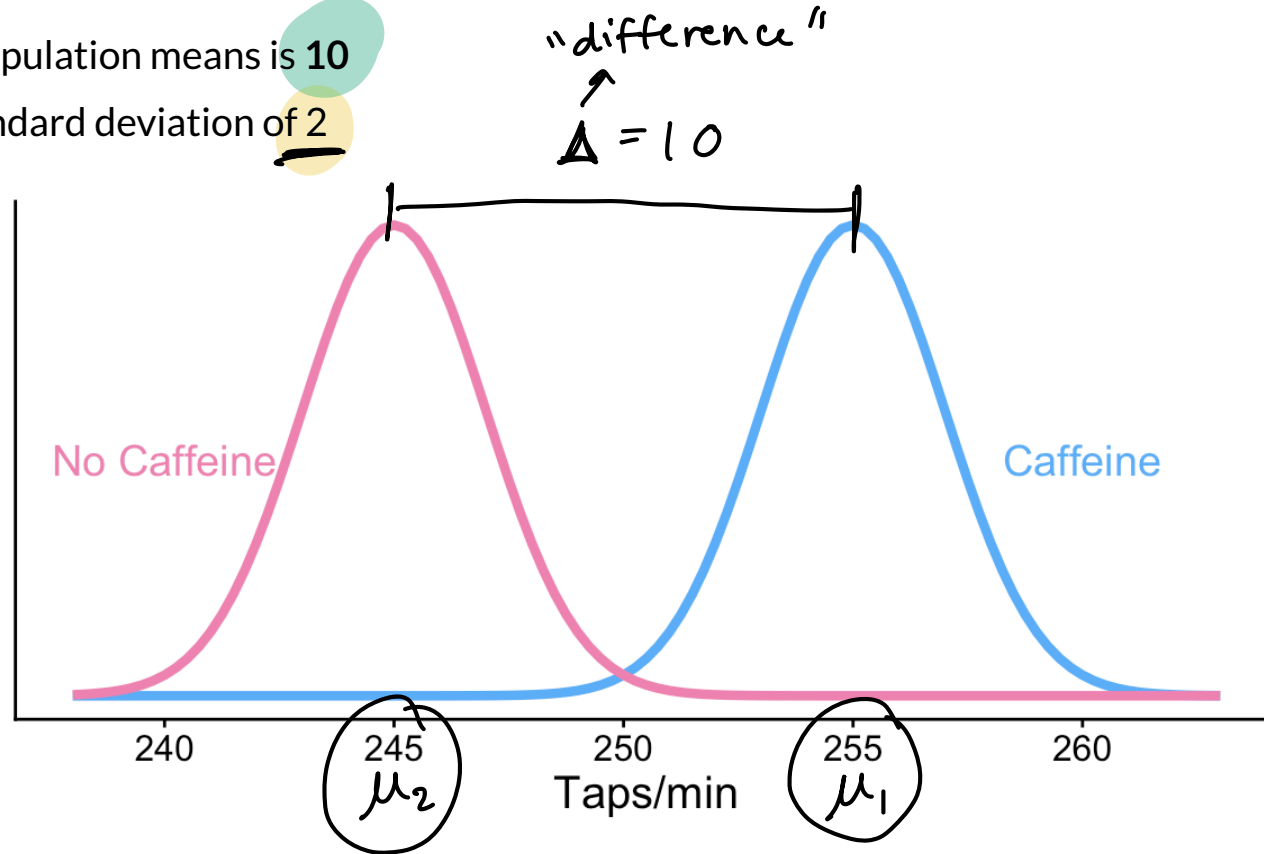
one way
to make
samples more
distinguishable
 $\uparrow n$
($\downarrow SE$)

- When we take two samples from these groups, do you think it would be easy to distinguish between the mean taps/min?
 - **Seems easier to distinguish here.** How did the standard deviation decrease?

sampling distrib:
 $\uparrow n$ $\downarrow SE$

What if the following were the true population distributions? Case 3

- Difference in population means is **10**
- Both have a standard deviation of **2**



- When we take two samples from these groups, do you think it would be easy to distinguish between the mean taps/min?

- Also seems easier to distinguish here

Handwritten note:

$\uparrow \mu_1 - \mu_2$ (not usually in our control)

There are a few things at play here

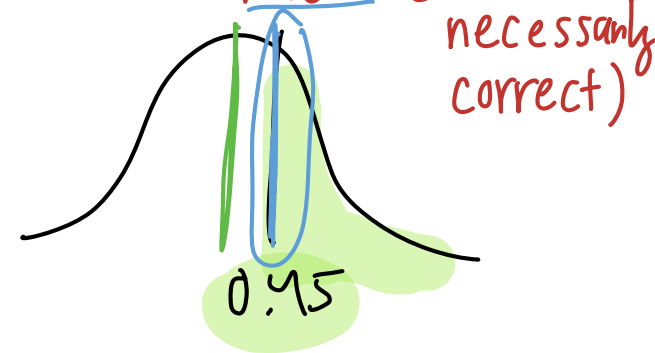
- There are several measurements that affect how easy it is to distinguish between two populations
- "Distinguish between two populations" = correctly reject the null hypothesis that they are the same

$$H_0: \mu_1 = \mu_2 \leftarrow \text{rejecting}$$

$$H_A: \mu_1 \neq \mu_2$$

- What elements are at play?

- Difference in population means $\mu_1 - \mu_2: \uparrow \text{diff} \Rightarrow \text{easier to distinguish}$
 - Number of samples from each population $\uparrow n \downarrow SE \Rightarrow \text{easier to distinguish}$
 - The **significance level** that we use for a cut off $\rightarrow \alpha = 0.05$ $\uparrow \alpha \Rightarrow \text{easier to reject null (not necessarily correct)}$
 - The **power** of our test
- More familiar with first two, but let's define #3 and #4 more

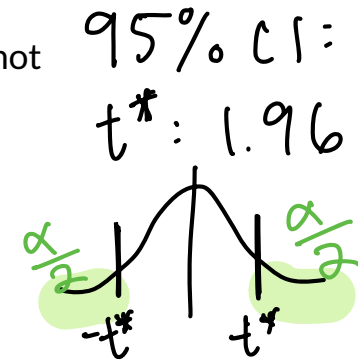


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Significance levels and critical values t^* @ a specific α

- **Critical values** are the cutoff values that determine whether a test statistic is statistically significant or not
 - Determined by the **significance level**
- If a test statistic is greater in **absolute value** than the critical value, we reject H_0



Critical Values for a Normal Distribution

$$\alpha = 0.05$$

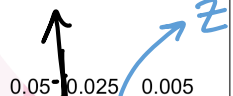
$$\left(\frac{\alpha}{2}\right) = 0.025$$

$$P(Z \geq z^*) = 0.025$$

$$P(Z \geq 1.96) = 0.025$$

crit value

$$z_x = 2.3$$



$2.3 > 1.96$
means I am
in my
rejection region

- We have been referring to critical values from the t-distribution as t^*
 - See how we calculate a specific confidence interval in Lesson 10

one-sided: $H_A: \mu > 0$
OR $H_A: \mu < 0$

13:49 Mon Nov 17

Join by Web **PollEv.com/nickywakim275**

$$H_0: \mu = 0$$
$$H_A: \mu \neq 0$$

$$\rightarrow \frac{\alpha}{2} = 0.05$$

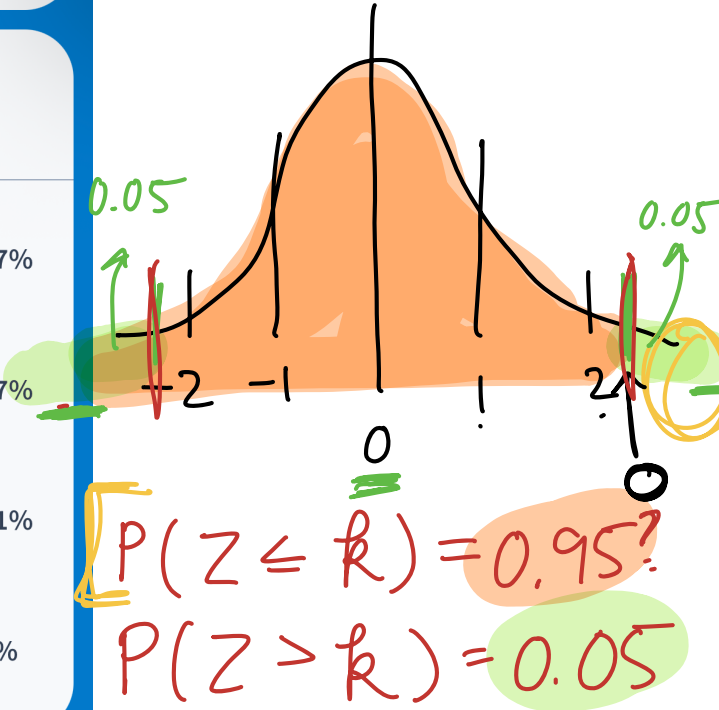
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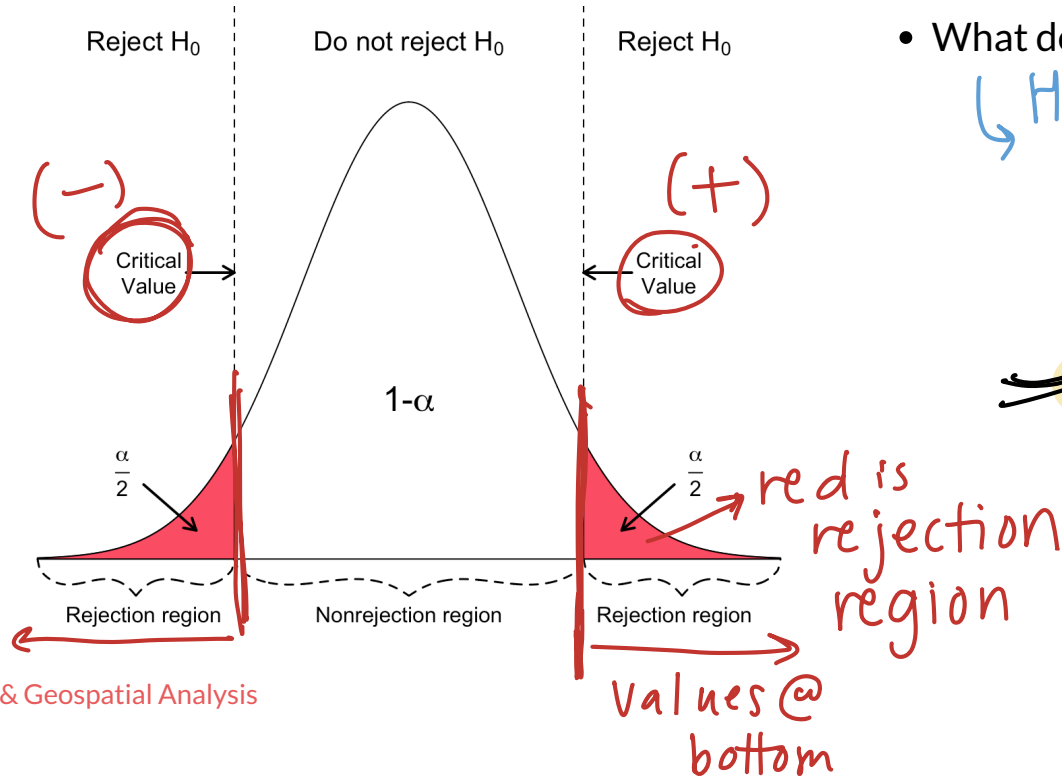
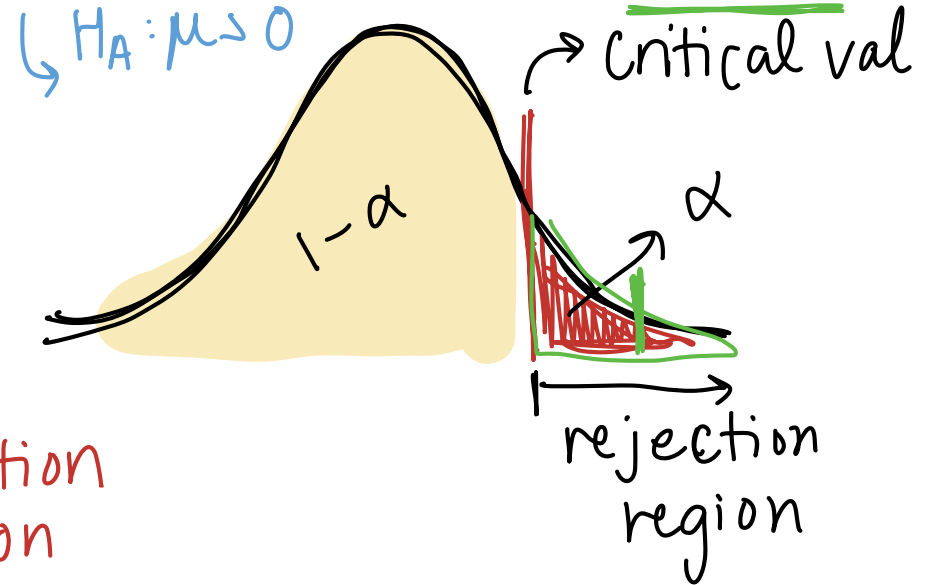
Rejection region, significance levels, and critical values

- If the absolute value of the test statistic is greater than the critical value, we reject H_0

- In this case the test statistic is in the **rejection region**.
- Otherwise it's in the non-rejection region.

↳ greater than critical val
OR less than the neg
critical value

- What do rejection regions look like for 1-sided tests?



Learning Objectives

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Let's start with some important definitions in words

- **Type I error (α)**: Probability of rejecting the null hypothesis given that the null is true
- **Type II error (β)**: Probability of failing to reject the null hypothesis given that the null hypothesis is false
- **Power (or sensitivity) ($1 - \beta$)**: Probability of rejecting the null hypothesis given that the null is false (correct)
- **Specificity ($1 - \alpha$)**: Probability of failing to reject the null hypothesis given that the null is true (correct)

		Our conclusion	
		<u>Fail to reject null hypothesis</u>	<u>Reject null hypothesis</u>
Underlying truth	Null hypothesis is true	Correct! (true negative) $1 - \alpha$	Type I error (false positive) probability = α
	Null hypothesis is false	Type II error (false negative) probability = β	power Correct! (true positive) $1 - \beta$

$$(1 - \alpha) + \alpha = 1$$

$$\beta + (1 - \beta) = 1$$

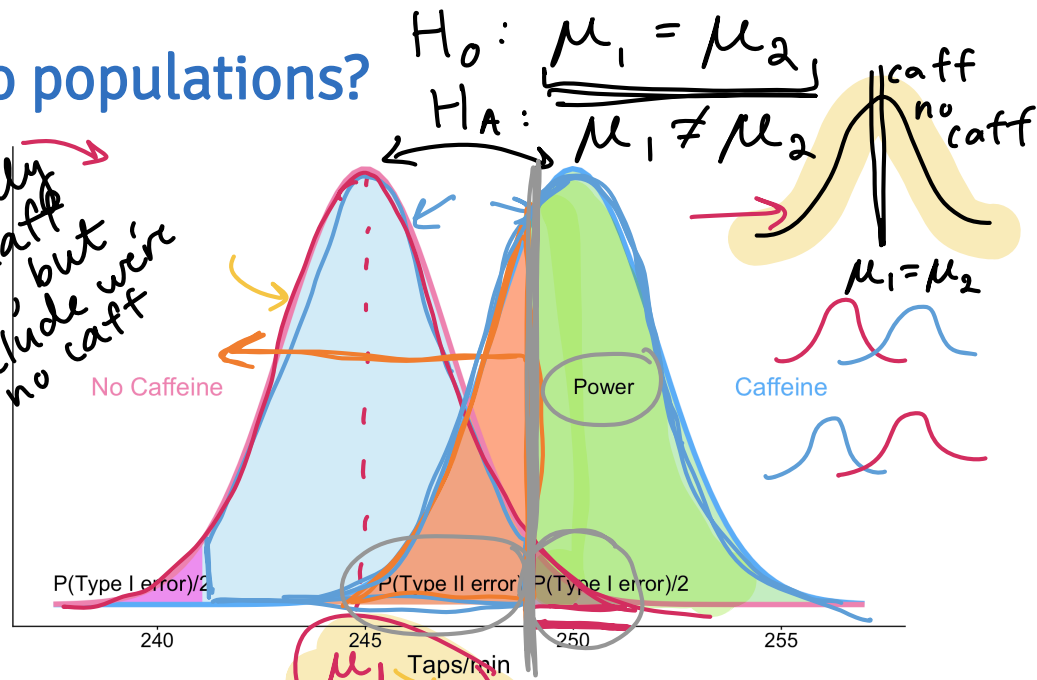
What does that look like with our two populations?

		Our conclusion	
		Fail to reject null hypothesis	Reject null hypothesis
Underlying truth	Null hypothesis is true	Correct! (true negative)	Type I error (false positive) probability = α
	Null hypothesis is false	Type II error (false negative) probability = β	Correct! (true positive) $1 - \beta$

two diff pop dist.

- α = probability of making a **Type I error**
 - This is the significance level (usually 0.05)
 - Set before study starts
- β = probability of making a **Type II error**
- Ideally we want
 - Small Type I & II errors $\downarrow \alpha \downarrow \beta$
 - Big power $\uparrow (1 - \beta)$

actually in caff dist, but conclude we're in no caff

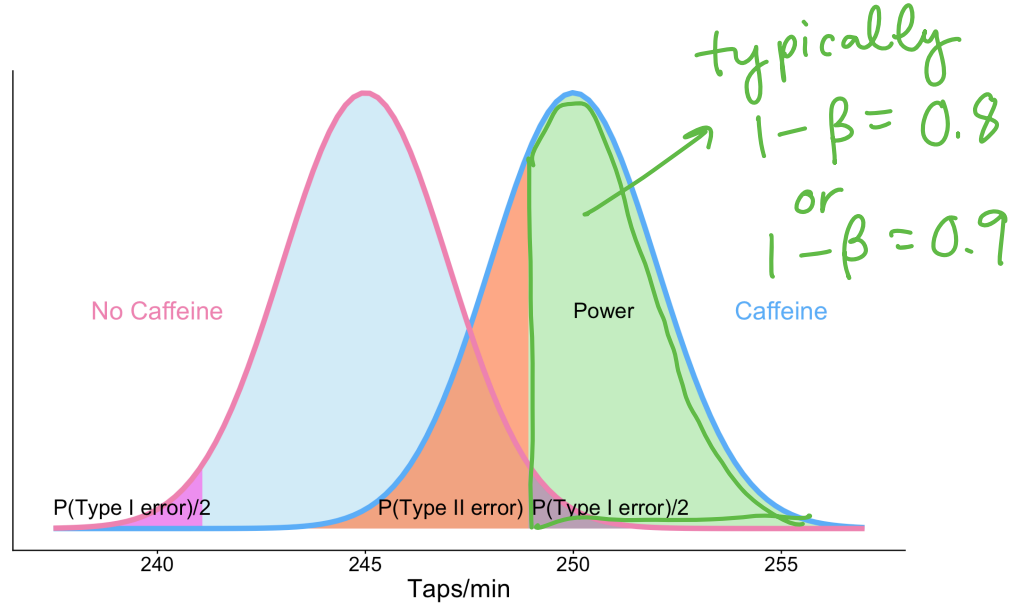


Power (or sensitivity) $(1 - \beta)$: Probability of rejecting the null hypothesis given that the null is false (correct)

- Power is the correct region that is usually in line with our study design: studies are often seeing if there is a distinction between two populations

Power

- **Power** (or sensitivity) ($1 - \beta$): Probability of rejecting the null hypothesis given that the null is false (correct)
- Power is also called the
 - true positive rate,
 - probability of detection, or
 - the *sensitivity* of a test
- Typically, we aim for 80% or 90% power



Let's demonstrate the relationship between error and power

From the applet at <https://rpsychologist.com/d3/NHST/>

Let's look at the following scenarios:

1. Solve for power: decreasing type 1 error (α)

2. Solve for power: increasing type 1 error (α)

- Takeaway: cannot minimize both type 1 and 2 error

3. Solve for power: decrease sample size

4. Solve for power: increase sample size

- Takeaway: increasing sample size increases power

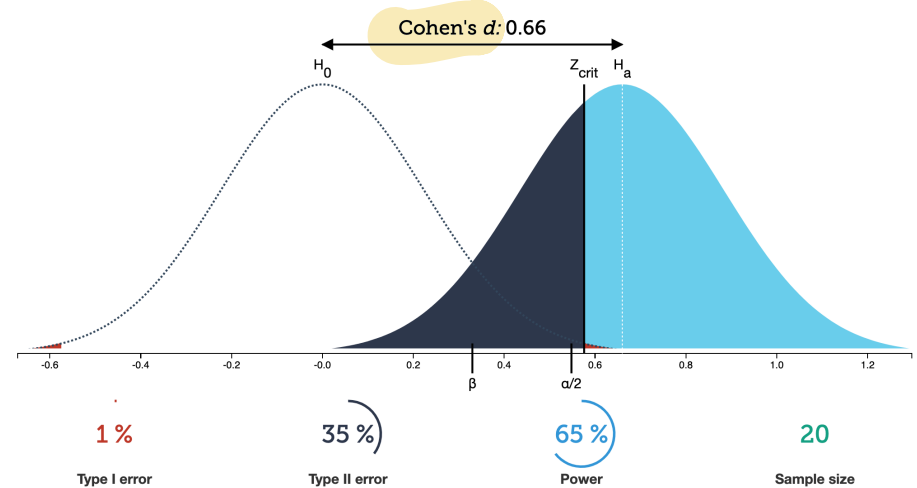
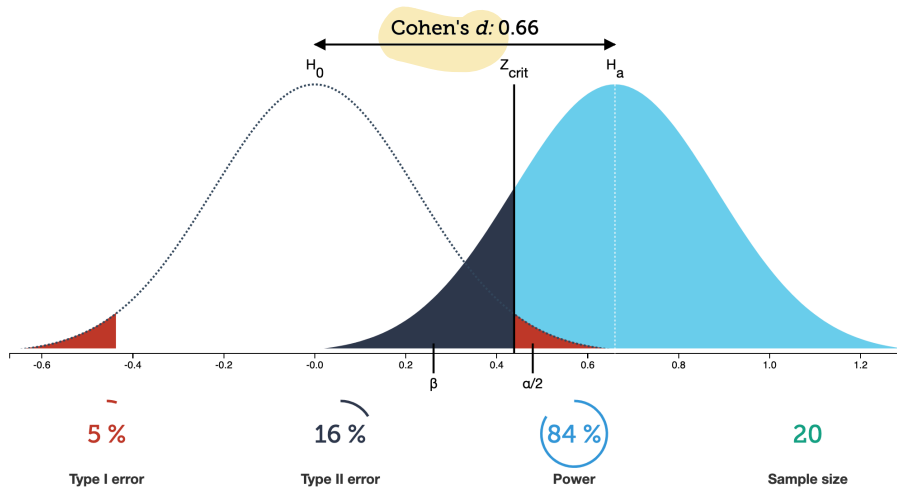
5. Solve for power: increase difference of means

6. Solve for power: decrease difference of means

- Takeaway: increasing difference in means increases power

If you want to keep revisiting these concepts!

From the applet at <https://rpsychologist.com/d3/NHST/>



- Cohen's d is just a standardized value to represent the difference in means:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$$

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Calculating power or sample size

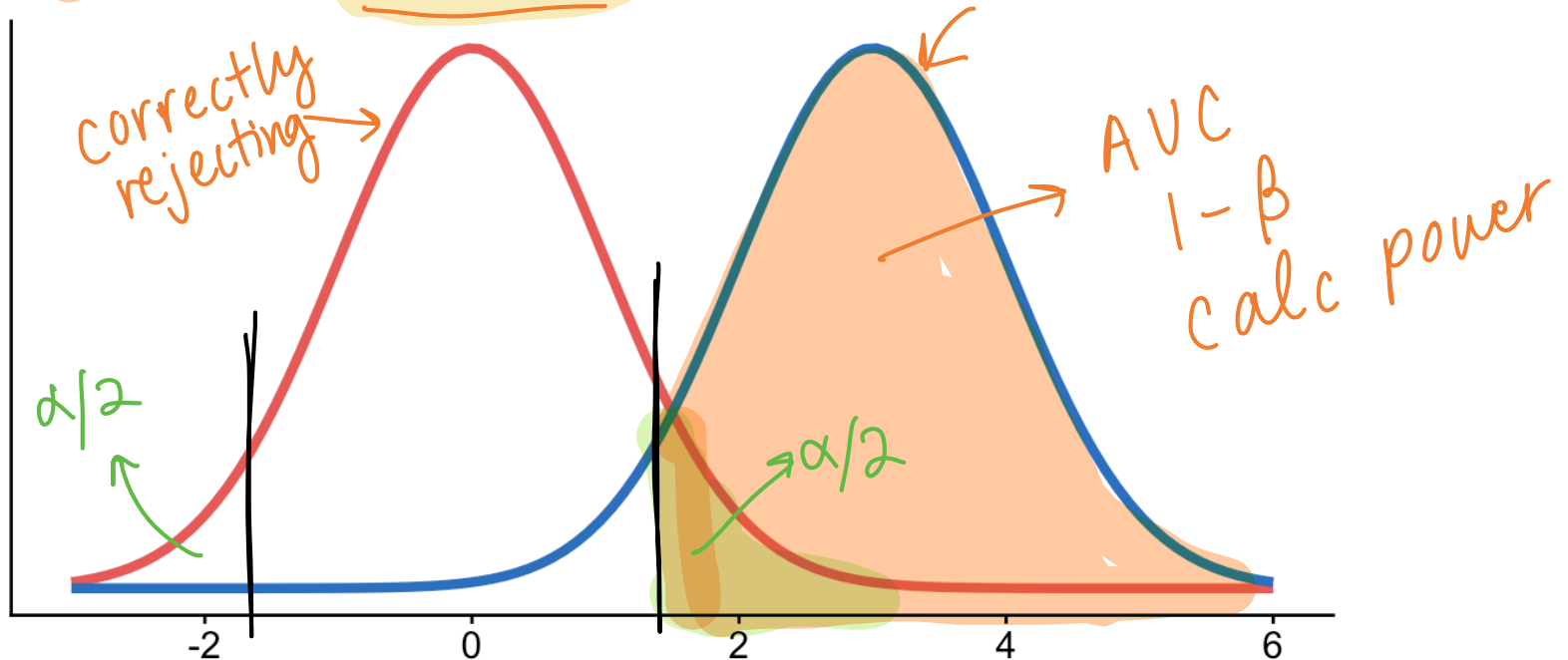
- Typically, before we set up a research study, we try to find the needed sample size to achieve 80% or 90% power
- If we have already have data, then we typically calculate the power based on the sample we have

Example calculating power (1/3)

Let's say we have:

- a **null population** with a normal distribution, centered at 0 with a standard deviation of 1 ($X_0 \sim \text{Norm}(0, 1)$)
- an **alternative population**, with normal dist'n, centered at 3 with standard deviation of 1 ($X_A \sim \text{Norm}(3, 1)$)

Find the power of a 2-sided test if the actual mean is 3 and our significance level is 0.05.



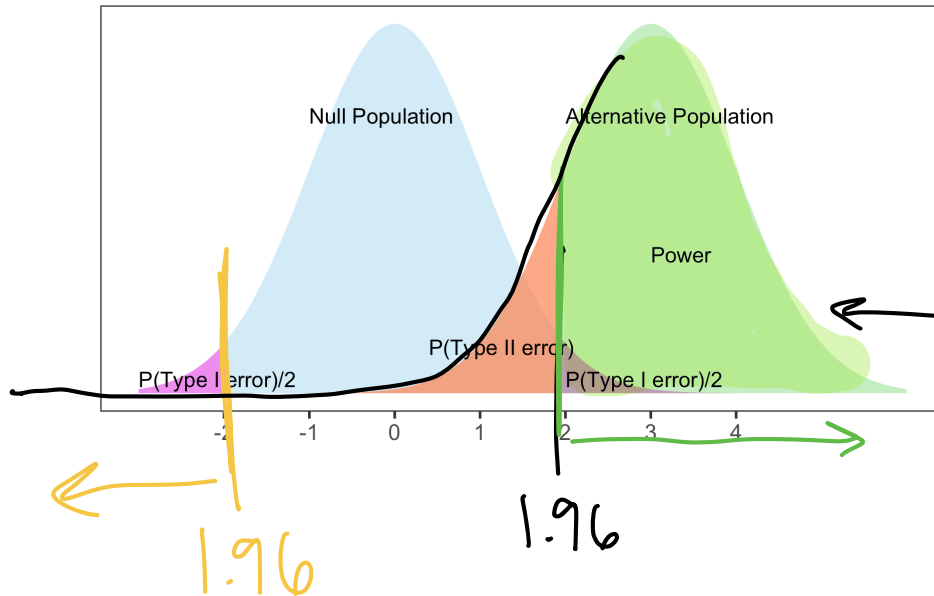
Example calculating power (2/3)

Let's say we have:

- a null population with a normal distribution, centered at 0 with a standard error of 1 ($X_0 \sim \text{Norm}(0, 1)$)
- an alternative population, centered at 3 with a standard error of 1 ($X_A \sim \text{Norm}(3, 1)$)

Find the power of a 2-sided test if the actual mean is 3 and our significance level is 0.05.

Type I & II errors and power

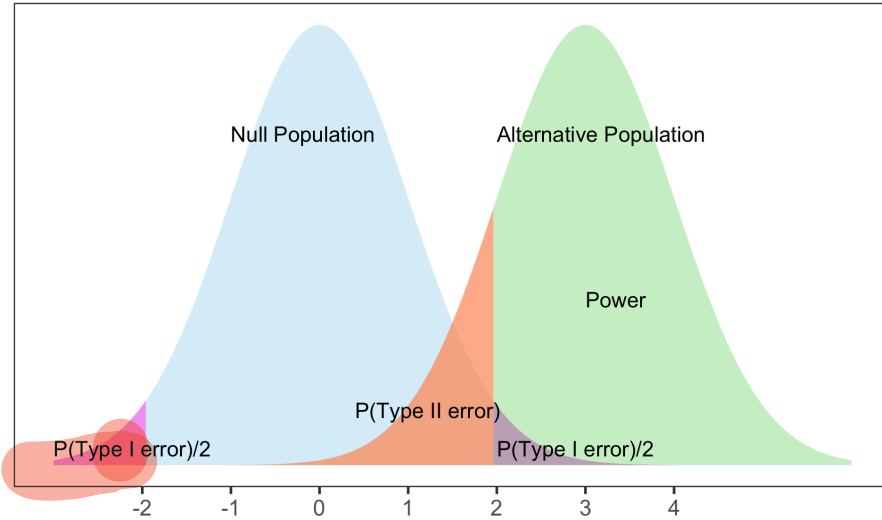


- Power = $P(\text{Reject } H_0 \text{ when alternative pop is true})$
 - Correctly reject null
- When $\alpha = 0.05$, we reject H_0 when the test statistic z is at least 1.96 (critical value is 1.96 under the null distribution)
- Then we need to calculate the probability that we are in the rejection regions given we are actually in the alternative population
- Thus under the alternative population, we need to calculate $P(X_A \leq -1.96) + P(X_A \geq 1.96)$

≈ 0

power

Example calculating power (3/3)



- Thus under the alternative population, we need to calculate $P(X_A \leq -1.96) + P(X_A \geq 1.96)$

- Under the alternative population we have

$$X_A \sim \text{Norm}(3, 1)$$

```
1 # left tail + right tail:  
2 pnorm(-1.96, mean=3, sd=1, lower.tail=TRUE) +  
3  
4 pnorm(1.96, mean=3, sd=1, lower.tail=FALSE)  
5
```

[1] 0.8508304

Answer: The power is 85%

- The left tail probability `pnorm(-1.96, mean=3, sd=1, lower.tail=TRUE)` is essentially 0 in this case.
- Note that this power calculation specified the value of the SE instead of the standard deviation and sample size n individually.

$$SE = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \frac{s}{\sqrt{n}}$$

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R package pwr for power analyses¹

- Use `pwr.t.test` for both one- and two-sample t-tests
- Specify all parameters *except* for the one being solved for

```
1 pwr.t.test(n = NULL,  
2           d = NULL, → diff b/w means  
3           sig.level = 0.05,  
4           power = NULL,  
5           type = c("two.sample", "one.sample", "paired"),  
6           alternative = c("two.sided", "less", "greater"))
```

- Leave out:
 - `n`: returns sample size
 - `d`: returns Cohen's d/effect size (next slide)
 - `sig.level`: get significance level (not typical)
 - `power`: returns power

What is Cohen's d ?

- d is **Cohen's d** effect size
 - Just a standardized way to measure the distance between the null mean and the alternative mean
- Examples of values: small = 0.2, medium = 0.5, large = 0.8

One-sample test (or paired t-test):

$$d = \frac{\bar{x} - \mu_0}{s}$$

$$d = \frac{\bar{x}_d - \delta_0}{s_d}$$

Two-sample test (independent):

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$$

- $\bar{x}_1 - \bar{x}_2$ is the difference in means between the two groups that one would want to be able to detect as being significant,
- s_{pooled} is the pooled SD between the two groups - often assume have same sd in each group

Sample size calculation for testing one mean

- Recall in our body temperature example that $\mu_0 = 98.6$ °F and $\bar{x} = 98.25$ °F.
 - The p -value from the hypothesis test was highly significant (very small).
 - What would the sample size n need to be for 80% power?
- Calculate n
 - Given α , power $(1 - \beta)$, "true" alternative mean μ , and null μ_0
 - Using calculated d :

$$d = \frac{\bar{x} - \mu_0}{s}$$

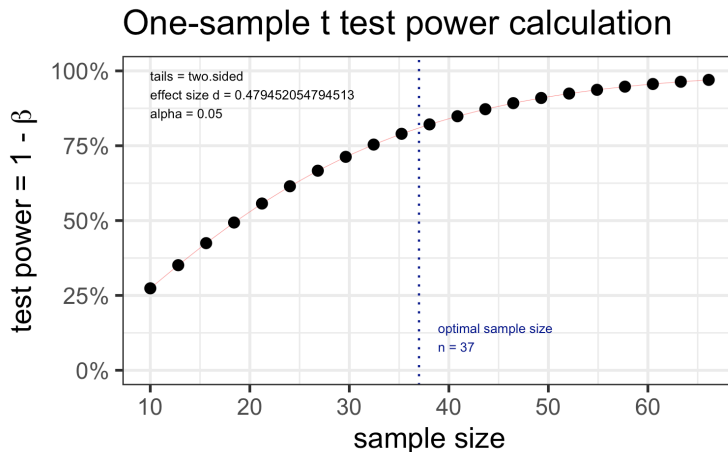
pwr: sample size for one mean test

Specify all parameters *except* for the sample size:

$$d = \frac{\bar{X} - \mu_0}{S} = \frac{98.25 - 98.6}{0.73}$$

```
1 library(pwr)
2 t.n <- pwr.t.test(
3   d = ((98.6-98.25)/0.73),
4   sig.level = 0.05, alpha = 0.05
5   power = 0.80,
6   type = "one.sample")
7
8 t.n
```

```
1 plot(t.n)
```



One-sample t test power calculation

```
n = 36.11196
d = 0.4794521
sig.level = 0.05
power = 0.8
alternative = two.sided
```

what we gave it

We need 37 individuals to detect this difference with 80% power.

round sample size UP

pwr: power for one mean test

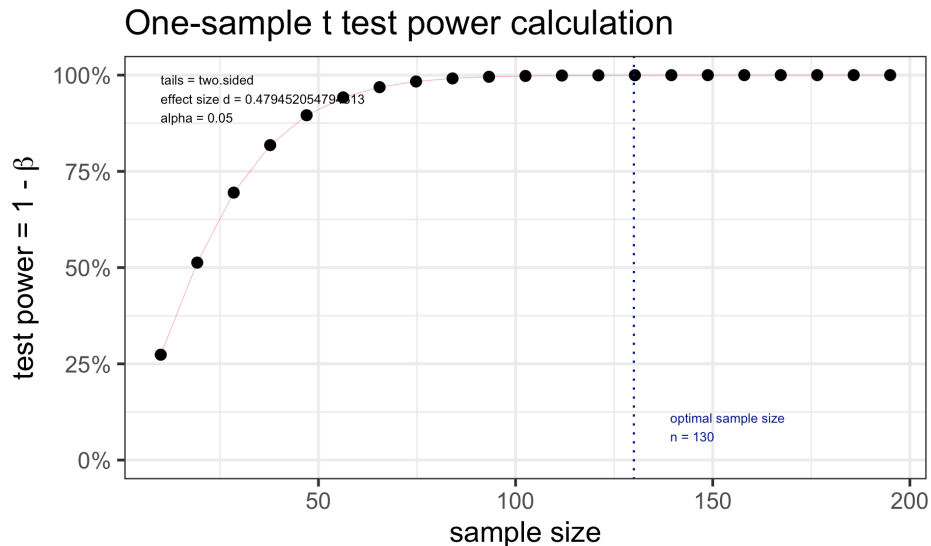
Specify all parameters *except* for the power:

```
1 t.power <- pwr.t.test(  
2   d = (98.6-98.25)/0.73,  
3   sig.level = 0.05,  
4   # power = 0.80,  
5   n = 130,  
6   type = "one.sample")  
7  
8 t.power
```

One-sample t test power calculation

```
n = 130  
d = 0.4794521  
sig.level = 0.05  
power = 0.9997354  
alternative = two.sided
```

```
1 plot(t.power)
```



We have **99.97% power** to detect this difference with 130 individuals.

pwr: Two-sample t-test: sample size

Example: Let's revisit our caffeine taps study. Investigators want to know what sample size they would need to detect a 2 point difference between the two groups. Assume the SD in both group samples is 2.6.

Specify all parameters *except for* the sample size:

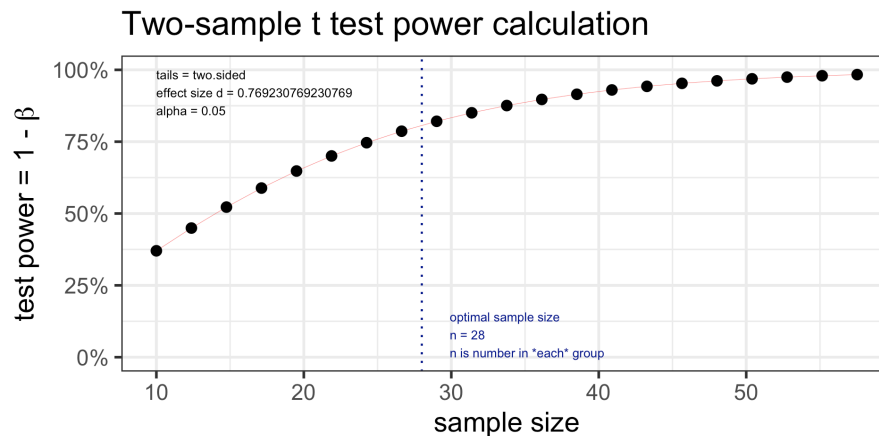
```
1 t2.n <- pwr.t.test(  
2   d = 2/2.6,  
3   sig.level = 0.05,  
4   power = 0.80,  
5   type = "two.sample")  
6 t2.n
```

Two-sample t test power calculation

```
      n = 27.52331  
      d = 0.7692308  
sig.level = 0.05  
  power = 0.8  
alternative = two.sided
```

NOTE: n is number in **each** group

```
1 plot(t2.n)
```



We need 28 individuals in each group to detect this difference with 80% power.

pwr: Two-sample t-test: power

Example: Let's revisit our caffeine taps study. Investigators want to know what power they have to detect a 2 point difference between the two groups. The two groups are both size 35 (like in our previous example). Assume the SD in both group samples is 2.6.

Specify all parameters *except* for the power:

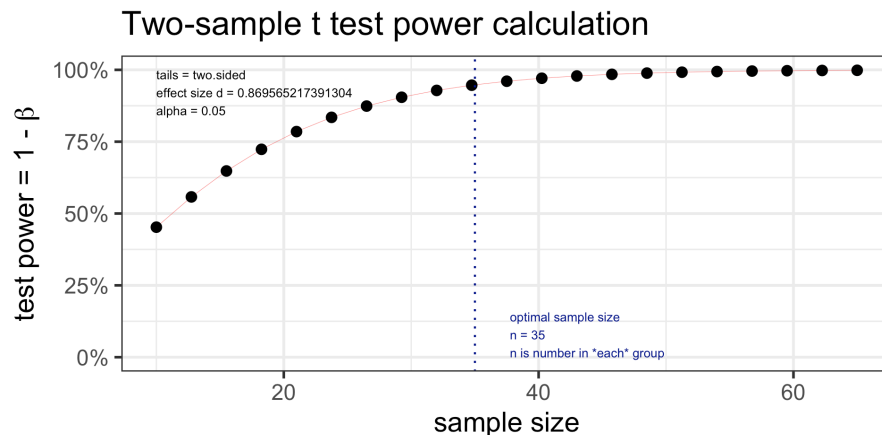
```
1 t2.power <- pwr.t.test(  
2   d = 2/2.3,  
3   sig.level = 0.05,  
4   n = 35,  
5   type = "two.sample")  
6 t2.power
```

Two-sample t test power calculation

```
n = 35  
d = 0.8695652  
sig.level = 0.05  
power = 0.9480091  
alternative = two.sided
```

NOTE: n is number in *each* group

```
1 plot(t2.power)
```



We have **94.8% power** to detect this difference with 35 individuals in each group.

Resources for power and sample size calculations

More software for power and sample size calculations: PASS

- PASS is a very powerful (& expensive) software that does power and sample size calculations for many advanced statistical modeling techniques.
 - Even if you don't have access to PASS, their **documentation** is very good and free online.
 - Documentation includes formulas and references.
 - PASS documentation for powering **means**
 - One mean, paired means, two independent means
- One-sample t-test documentation: https://www.ncss.com/wp-content/themes/ncss/pdf/Procedures/PASS/One-Sample_T-Tests.pdf

OCTRI-BERD power & sample size presentations

- **Power and Sample Size 101**
 - Presented by Meike Niederhausen; April 13, 2023
 - Slides: <http://bit.ly/PSS101-BERD-April2023>
 - [Recording](#)
- **Power and Sample Size for Clinical Trials: An Introduction**
 - Presented by Yiyi Chen; Feb 18, 2021
 - Slides: <http://bit.ly/PSS-ClinicalTrials>
 - [Recording](#)
- **Planning a Study with Power and Sample Size Considerations in Mind**
 - Presented by David Yanez; May 29, 2019
 - [Slides](#)
 - [Recording](#)
- **Power and Sample Size Simulations in R**
 - Presented by Robin Baudier; Sept 21, 2023
 - [Slides](#)
 - [Recording](#)